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The content extracted is as follows:

Therefore, it calls for a renewal of the firm-boundary problem and the very definition of the firm.

In other words, platform economics paradoxically demonstrate more than traditional firms that the capitalist subjection of labor relies entirely upon the economic power of the owners of the means of the production which are at the source of new forms of cooperation between producers, consumers, users, et cetera, on which its business models rely (Bauwens, Mendoza, and Iacomella 2012).

By reactivating the firm-boundary problem, the article will show how labor appropriation steers firms' decisions concerning integration.

The firm is a technological center since the relation of production relies upon and is carried out by a productive and organizational technology.

By stressing the relation of production between the owners of the means of production and the direct producers, the approach of the firm supported here should allow us to cover the different existing models of the firm, from the classical firm to hybrid models, around which the boundary debate has revolved, to digital platforms.

In order to emphasize this aspect common to the different kinds of firms, I will show how the definitions of the firm by transaction-cost economics (TCE), property-rights theory (PRT), and resource-based theory (RBT) encounter the firm-boundary problem and how this problem gets resolved by putting the exploitation of social labor at the heart of integration strategies of the firm.

After having recalled the history of the firm-boundary problem, emphasis will be put on the way it has been dealt with by TCE, then PRT and finally RBT, in accordance with their respective approaches of the firm.

The analysis of quasi-firms, in particular the Toyota quasi-integration system, will show that only a definition of the firm that deals with labor as both producing goods and services and creating surplus value appropriated as profits may account for hybrid models while drawing a clear boundary between firm and market.

Digital Platforms and the Theory of the Firm

Moreover, crowdsourcing blurs the boundaries of the firm not only because it uses collaboration mass weapons (Tapscott and Williams 2010) extended to the whole of society, but also because local independent workers potentially have access to the global market.

In digital platforms, as well as in traditional firms, the manager's power over the worker relies entirely upon the power to manage the means of production necessary to the workers' social and individual reproduction.

By founding the authority and incentives on property, PRT not only makes no distinction between firm and market except in accordance to their degree of property but also neglects the fact that the property of non-human assets determines a relation of production exceeding the framework of interpersonal property rights.

In the capitalist firm in general, the property rights ordering the relations between managers and employees are structured not, as PRT claims, by the collective use of the nonhuman assets as such, but by the social relation of production under which these non-human assets are used and valorized. Since the use and valorization of non-human assets are planned through contractual and organizational arrangements, we suggest defining the firm as a techno-institutional center of valorization (i.e., of extraction of surplus labor).

By focusing on the way labor is organized and extracted in order to valorize capital, this approach of the firm may renew and clarify the problem of the firm's boundaries.

The Coasian Genesis of the Firm-Boundary Problem

His question implies that standard theory only dealt with owners of goods, services or assets, and that production could be de jure totally realized through market exchange, as

in the putting-out system.

Coase's theory of the firm therefore does not deal with the origins of production under capitalism, but directly starts with the market economy whose failures push the producers or asset owners to set up mechanisms of coordination not covered by market coordination: In the beginning, there were markets, said Williamson (1975, 20).

But since the basic assumptions of standard theory are preserved, notably methodological individualism and preference maximization, TCE assumes a vision of the firm primarily as a bundle of bilateral exchange relations, not as a proper institutional space of production organization.

The growth of large corporations, the power of firms to control prices and markets, the scientific management of labor and production within multidivisional firms, et cetera, stress the fact that firms control the markets rather than reacting to their signals (Chandler 1977).

The Fuzzy Boundaries of the Firm in Transaction-Cost Theory

The firm-boundary problem arose again and developed from the 1980s onwards, when liberalization and deregulation of financial markets launched the second wave of mergers and acquisitions, while financial analysts serving institutional investors forced large corporations to fall back on their core competency and outsource numerous parts of the value chain (Prahalad and Hamel 1990).

Hence the renewed interest in TCE, which gives theoretical tools to the firm's choice to make or buy and sees the firm and its boundaries in terms of exchange relations, not only between managers and employees, but between the firm and its suppliers, subcontractors, customers, et cetera (Williamson 1975).

From this viewpoint, the boundary problem was closely related to the hold-up problem arising as soon as firms make investments in specific assets (Klein, Crawford, and Alchian 1978).

According to TCE, what distinguishes the firm from the market and justifies it is not the specific nature of the firm's activity (production) but the contractual relationship specific to this activity (the employment relation).

Nevertheless, though TCE gives a theoretical framework to integration strategies, the boundaries of the firm seem to remain theoretically ill-defined. Why would savings in transaction costs not lead to the integration of the entire production by one giant firm?

This problem emphasizes the failure of TCE to understand the firm as a process of production distinct from exchange relations rather than just in accordance with the market (i.e., only as a solution to market failures).

Not only does TCE embrace the manager's point of view by theorizing her choices and trade-offs, but it also grasps her activity as exclusively concerned with the profits of the firm and disconnected from the management of the production process.

Since PRT's critique of TCE is based on the requirements of the combination of human and non-human assets necessary to the productive activity and thus to the valorization process, let us look at the way it solves the boundary-problem of the firm.

Property Rights Theory's Critique of Transaction-Cost Theory and its Failure to Solve the Integration Problem

PRT's vision of the firm comes closer to its productive purpose by linking the manager's role to ownership of tradable assets, and thus by seeing the relations within the firm as social relations of production.

As defined by Alchian and Demsetz (1972), the firm is a team production in which the combination of assets to produce output requires the control of productivity and distribution of incomes between human assets by a manager emerging from the team.

What characterizes the capitalist firm is joint input production with numerous owners, among whom one (the manager) is common to all the contracts of the joint inputs and holds the residual claim as an incentive to monitor the assets.

Therefore, he defines the firm with Sanford Grossman as an entity owning means of production (Grossman and Hart 1986, 692) and argues that ownership permits holding all the rights unspecified by the contract (i.e., the residual rights).

If we do not dismiss the totally abstract neoclassical assumption of maximization under constraint of decreasing returns to scale (Williamson 1985, 93), trade-off between internal production costs and market transaction costs theoretically sets no limit to the expansion of the firm.

Whereas PRT allows an explanation of how assets are distributed between firms, given a set of assets and investment activities, it fails to explain how these property rights are created. In order to grasp it and thus account for the very existence of the firm and its boundaries, it is more important to explain how and why investment activities are themselves distributed. According to Holmström, the boundary question is what the firms do rather than what they possess (Holmström and Roberts 1998, 87).

Nonetheless, this move from assets to activities brings out new questions: why do firms (or managers) own mostly non-human assets used in production?

Put differently, why does the capitalist firm exist, insofar as it is based on the fact that workers do not own their means of production?

More generally, although Holmström and Roberts show the benefits of owning means of production and thus of making rather than buying, they do not explain the conditions allowing firms to own these means and take advantage of them by employing human assets deprived of these means.

Moreover, as will be shown, control and incentives over human assets do not allow one to delimit the boundaries of the firm, any more than residual rights of control do.

However, by emphasizing the importance of incentives and efficiency of human assets in the ownership of non-human assets, Holmström and Roberts have shifted the focus from assets to activities and thus from the firm's structure of property to its joint process of production and valorization.

The Limitless Boundaries of the Resource-Based Firm

In fact, drawing the boundaries of the capitalist firm requires knowing its nature and function.

If one understands capital as a mode of production and not just as a source of revenue, as Thomas Piketty (2014) contends, the function of the firm appears as dual and consists in producing goods and services in order to valorize productive investment (i.e., to make profits from investment).

The firm is then a techno-institutional center in which production and distribution of goods and services by labor are the means of valorizing the assets used in that process.

Prima facie, RBT seems to be the only firm model to explicitly link valorization to the production of specific goods or services by defining the firm as a bundle of capabilities, organizational routines, and knowledge produced thanks to a combination of resources and cooperation between stakeholders of the firm (Coriat and Weinstein 2010, 60-62).

Furthermore, although Pitelis and Teece claim to be able to grasp the positive nature of the firm in opposition to TCE, which justifies it only negatively as a solution to market failures, their approach does not solve the firm-boundary problem, insofar as it extends to all the relations of the firm, and thus to the entire economy, the power of creating the value captured by the firm (Pitelis and Teece 2009, 10).

How can one delimit the boundaries of a firm whose management is based on communities of users or sharers connected to worldwide communication networks such as digital platforms?

For different reasons, neither TCE and PRT, nor RBT are able to strictly delimit the boundaries of the firm: TCE and RPT because the firm differs from market exclusively with respect to the nature and amount of costs resulting from the incomplete contracts; RBT because the firm is not distinguished from the whole of its connections with all its stakeholders.

But even when the concept of capability is strictly limited to the organization of production, the boundaries of the firm remain undefined. According to RBT, the choice to integrate one or some parts of the value chain depends on the possibility to develop a specific competence (Argyres and Zenger 2012).

Moreover, the indeterminate boundaries of the resource-based firm seem to reinforce the position of TCE regarding this issue.

Indeed, Nicolai Foss has remarked that the boundaries of the firm are difficult to rationalize except on a standard contractual basis (Foss 1996, 475) when firms have the opportunity to communicate knowledge to supplier firms (Dyer 1997, 546).

If the firm is defined as a bundle of organizational capabilities, therefore, its boundaries would be delimited by all the assets contributing both to the production of routines and competences and to the process of valorization.

It would be therefore necessary to exclude from the boundaries of the firm the suppliers, subcontractors, customers, et cetera who do not take part in the valorization process of productive assets; indeed the economic operations between them and the firm consist in exchange of money for goods or services, notwithstanding information exchange

contributing to the creation of the contracting firm's specific resources.

Thus, the concept of capability or resource is either too narrow or too broad for drawing the boundaries of the firm: too narrow, since it excludes the valorization process from the production process; too broad, since some resources are created through relations of exchange with parties external to the contracting firm and are not part of its own valorization process (Langlois 2003).

It fuses production relations and market relations under the same concept, even though a clear distinction between them is necessary to delimit the boundaries of the firm within which the valorization process takes place.

The Quasi-Firm-Boundary Problem

We have seen how TCE, PRT, and RBT meet difficulties in dealing with vertical integration and firm-boundary problems.

In addition to governance structure and transaction costs, the characteristics of resources determine the boundary choice of a firm (Madhok 2002, 541).

To the extent that inter-firm collaboration permits joint production of resources, the distinction between production and exchange is blurred since firms choose hybrid models in order to not only lower their transaction costs but to increase efficiency and create more value.

But since the distinction between production and exchange is blurred, how can we delimit the boundaries of the firm?

Here again, the quasi-firm issue is dealt with in terms of benefits for firms to adopt hybrid forms, yet by obfuscating the boundary problem arising with these forms.

Michael Jacobides and William Billinger show through an empirical study that the boundary problem does not arise from the distinction between buying or making, but that firms may choose mixed and dynamic forms between them (Jacobides and Billinger 2006, 258).

Even though their study shows how a firm is led to change its boundaries over time according to valorization strategies, it does not deal with the question: what are the boundaries of a firm engaging in hybrid relations with other firms?

But it does not deal with the boundary problem arising from these hybrid forms in which the distinction of production and exchange is blurred. Interestingly, Mark Granovetter (1985) sees the quasi-firm as closer to vertical integration than to market relations but understands it as interactions between a contractor and subcontractors geographically close to each other where the latter are supervised by the former (Granovetter 1985, 497-498).

I claimed above that, since subcontracting is a market relation between two or more legally independent firms, it should exclude subcontractors from the boundaries of the contracting firm.

However, what justifies exclusion of subcontractors from the boundaries of the firm is not the recourse to the market as such, but the fact that the value exchanged between two independent firms does not produce more value than previously created by these firms.

However, the subcontractor, despite its legal independence, must be seen as included within the boundaries of the contracting firm.

Indeed, in a great number of industrial strategies, recourse to market is merely formal and does not relate to incentives or transaction cost problems, but only to productive efficiency or combination and organization of the value chain. On the other hand, quasi-integration concerns that concepts of capability or resource, insofar as they are sources of rents, no longer allow us to delimit the boundaries of the firm.

Here, the firm's strategy consists simultaneously of concentrating on its core competencies and appropriating the surplus value produced in the outsourced segments of the value chain. The specific competence of the firm is neither sufficient to define the nature of the firm nor to delimit its boundaries.

The Toyota system illustrates a case where the firm is grasped, beyond its traditional boundaries, by its specific competencies.

But it is only insofar as these competencies concern the process of production, and thus of valorization, that RBT can delimit the boundaries of the Japanese firm more accurately than TCE or PRT.

RBT therefore not only identifies the firm differently despite having similar empirical results as TCE (Carter and Hodgson 2006) but, as we have seen, can indefinitely extend the boundaries of the firm as long as relational capital is not related to the processes of production and valorization of the firm's productive assets.

The Japanese firm and its suppliers and subcontractors, as integrated from the perspective of their organizational and technical competencies on the one hand, and contractualized from the perspective of their formal relations on the other, are indeed quasi-integrated and together draw the boundaries of a single firm.

In their seminal article on the boundaries of the firm, Klein, Crawford, and Alchian claim that hold-up threatens not only the supplier or subcontractor which has invested in specific assets, but also the contracting firm, since the subcontractor may hold a monopolistic position vis-à-vis the firm (Klein, Crawford, and Alchian 1978, 299).

But how, practically, could a subcontractor, whose assets are totally dedicated to the production of output exclusively bought by the contracting firm, ever exert any pressure on the latter?

It basically comes under the logic of the valorization allowed by new technological possibilities (NTICs, polyvalent machinery, modular production, et cetera).² Hybrid models such as quasi-integration challenge traditional theories of the firm.

Moreover, they show how the joint logic of production and valorization processes explains the nature and stakes of the relationship between contracting firms and delimits clearly their boundaries which remain otherwise blurred. Although contractual relations apparently prevail among the multiple stakeholders, hybrid models show paradoxically the importance of living labor exploitation in the understanding of industrial strategy of the firms.

The Digital Platform-Boundary Solution

We have seen how TCE, PRT, and RBT have faced difficulties in solving the (quasi) firm-boundary problem.

TCE and from this point of view also PRT clearly distinguishes the firm from the market, but only from the perspective of their contractual relations, not with respect to the specific function of the firm regarding the market: production of goods or services incorporating surplus value appropriated as profits.

Conversely, RBT understands the firm as an institution creating routines and knowledge dedicated to specific goods or services, yet confuses production of surplus value with creation of competitive advantage (which is supposed to be the source of rents or quasi-rents).

Thus, according to this approach of the firm as a bundle of resources giving rise to organizational routines or specific knowledge, rents and quasi-rents arise as much from advantages drawn from market relations with its customers, suppliers, et cetera as from efficiency of the production process or specificity of the product.

Strangely, each of these approaches fails to draw the boundaries of the firm clearly, being unable to distinguish adequately the firm from the market: TCE and PRT, because they bring the contractual nature of the specific market relation into the firm, even though PRT in particular admits the co-operative nature of production; and RBT because it does not draw any clear distinction between the advantages of the firm drawn from its market relations and those resulting from the organization of production.

But if digital platforms may relax direct authority and control, they nevertheless share with all other kinds of capitalist firms the relation of production based on the property and use of non-human assets by capitalists who extract living labor in order to valorize these social means of production.

This relation of production suggests seeing the firm as a techno-institutional center of extraction of surplus labor.

The focus on the private economic power on social production allows us to distinguish effectively between firm and market whatever is the form of hybridization between them.

Concluding Remarks

Since integration does not delimit the boundaries of the firm, solving the firm-boundary problem depends on a clear understanding of the nature and function of the firm.

Defined as such, the capitalist firm includes digital platforms, where production of value is fragmented into a large number of individual provisions of services, since it frees the firm-boundary problem from governance constraints on transactions as well as organizational constraints on knowledge and resources, in the same way as ICTs have practically freed the firm from those same constraints.